REPORT

Library Management System Project Report

1. Introduction

A Library Management System is a software-based solution designed to manage and organize the daily activities of a library in a systematic and efficient manner. In traditional libraries, all records are maintained manually in registers, which can be time-consuming, difficult to maintain, and prone to human errors. Searching for a particular book, updating records, or managing issued books becomes a challenging task when handled manually.

This project aims to overcome these problems by developing a simple Library Management System using the C programming language. The system allows the user to perform important operations such as adding new books, displaying all available books, searching for a specific book, issuing books, returning books, and deleting book records.

The project is designed using fundamental programming concepts such as structures, pointers, linked lists, and dynamic memory allocation. Instead of using arrays with fixed sizes, this system uses linked lists, which provide flexibility and efficient memory usage. This means the system can handle any number of books without wasting memory.

The main goal of this project is not only to create a working system but also to understand how programming concepts are applied in real-life situations. It helps students build a strong foundation in C programming and data structures.

2. Objective

The primary objective of this project is to develop a user-friendly and efficient system that can manage library records effectively. This project focuses on improving both theoretical knowledge and practical skills.

The key objectives of this project are as follows:

- To understand and implement dynamic memory allocation in C programming.
- To learn how linked lists work and how they can be used in real-world applications.
- To perform basic data operations such as insertion, deletion, searching, and updating records.
- To design a menu-driven program that is simple and easy to use for users.
- To improve logical thinking and problem-solving abilities.
- To understand how data can be organized and managed efficiently.

This project also helps students gain confidence in writing programs and understanding how software systems function.

3. Tools and Technology Used

This project is developed using simple and commonly used tools and technologies, which makes it easy to understand and implement.

Programming Language: C

The C programming language is used because it is powerful, fast, and widely used for system-level programming. It also helps in understanding low-level concepts like memory management.

Software: CodeBlocks / Dev C++

These software tools are used to write, compile, and execute the C program. They provide an easy interface for coding and debugging.

Platform: Windows

The program is designed to run on Windows operating systems, which are commonly used by students.

Concepts Used in the Project:

- Structures (to store book details)
- Pointers (to link nodes)
- Linked Lists (to store dynamic data)
- Dynamic Memory Allocation (using malloc and free functions)

These technologies together help in building a functional and efficient system.

4. Methodology

The system is developed using a linked list data structure. A linked list consists of multiple nodes, where each node stores data and a pointer to the next node.

In this project, each node represents a book and contains the following details:

- Book ID
- Book Title
- Author Name
- Book Status (Available or Issued)
- Pointer to the next book

Whenever a new book is added, memory is allocated dynamically using the malloc() function. This ensures that memory is used only when required and there is no limitation on the number of books.

When a book is deleted, the memory is freed using the free() function, which helps in efficient memory management.

The program follows a step-by-step process:

1. The program starts and displays a menu.
2. The user selects an option.
3. The program performs the selected operation.
4. The result is displayed.
5. The process repeats until the user exits.

This approach makes the system simple, flexible, and easy to understand.

5. Features of the System

The system provides multiple features that help in managing library records efficiently.

Add New Book

This feature allows the user to add a new book to the system. The user enters details such as book ID, title, and author name. A new node is created, and the book is added to the linked list. This operation increases the size of the list dynamically.

Display All Books

This feature displays all the books stored in the system. It shows complete details such as ID, title, author, and status. This helps the user to view all available records in one place.

Search Book

The search feature allows the user to find a book using its ID. The system checks each node in the linked list and displays the book details if found. If the book is not found, an appropriate message is shown.

Issue Book

This feature is used to issue a book to a user. When a book is issued, its status changes from “Available” to “Issued”. This helps in tracking which books are currently in use.

Return Book

When a user returns a book, this feature updates the status from “Issued” to “Available”. This ensures that the book becomes available for others.

Delete Book

This feature allows the user to remove a book from the system. The corresponding node is deleted, and memory is freed. This helps in maintaining clean and updated records.

6. System Working

The working of the system is simple and interactive. When the program starts, it displays a menu with different options (as shown in the output section of your report on page 4).

The user selects an option by entering a number. Based on the selected option:

- If the user selects option 1, the system asks for book details and adds a new book.
- If the user selects option 2, all books are displayed.
- If the user selects option 3, the system searches for a book.
- If the user selects option 4, a book is issued.
- If the user selects option 5, a book is returned.
- If the user selects option 6, a book is deleted.
- If the user selects option 7, the program exits.

The system keeps running in a loop until the user decides to exit.

7. Data Structure Used

The main data structure used in this project is a linked list. A linked list is a collection of nodes where each node contains data and a pointer to the next node.

Unlike arrays, linked lists do not require a fixed size. This makes them more flexible and memory-efficient.

Advantages of Using Linked List:

- Dynamic size (can grow or shrink)
- Efficient insertion and deletion
- Better memory utilization
- No wastage of memory

Importance in This Project:

The linked list is important because it allows the system to handle an unlimited number of books. It also makes operations like adding and deleting books easier and faster.

8. Advantages of the System

This system provides several advantages:

- Easy to understand and use
- Reduces manual work
- Saves time and effort
- Uses memory efficiently
- No fixed size limitation
- Helps in learning important programming concepts

It is a good project for beginners and students.

9. Limitations of the System

Despite its usefulness, the system has some limitations

:

- No graphical user interface (only text-based)
- Data is not saved permanently after closing the program
- Limited features compared to real systems
- Not suitable for large-scale libraries

These limitations can be improved in future versions.

10. Future Scope

The system can be improved in many ways:

- Add file handling to store data permanently
- Develop a graphical user interface (GUI)
- Add login and authentication system
- Include student and staff records
- Add fine calculation for late returns
- Convert into a web-based or mobile application

These improvements will make the system more advanced and practical.

11. Output Explanation

The output of the system clearly shows how the program works (as shown in page 4 and 5 of your report).

- The menu displays all available options.
- When adding a book, the system confirms successful addition.
- Display shows all book records in a structured format.
- Search shows the book details if found.
- Issue and return update the book status.
- Delete removes the book from the system.

These outputs prove that the system is working correctly.

12. Conclusion

This project successfully demonstrates how a Library Management System can be implemented using the C programming language. It helps in understanding important concepts such as linked lists, pointers, and dynamic memory allocation.

The project is simple but effective and provides a strong base for developing more complex systems in the future. It also improves programming skills and logical thinking.

Overall, this project is very useful for students and beginners who want to learn how real-world systems are built using programming

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OUTPUT

```
--- Library Menu ---
1. Add Book
2. Display Books
3. Search Book
4. Issue Book
5. Return Book
6. Delete Book
7. Exit
Enter choice:
```

1. **ADD a Book**

```
Enter choice: 1
Enter ID: 01
Enter Title: Gunaho Ka Devta
Enter Author: Dharamvir Bharti
Book added
```

2. **Display BOOK**

```
nter choice: 2

D       Title           Author          Status
        Gunaho Ka Devta                 Dharamvir Bharti                Available
```

3. **Search Book**

```
Enter choice: 3
Enter ID: 01
Found: Gunaho Ka Devta  by Dharamvir Bharti (Available)
```

4. **Issue Book**

```
Enter choice: 4
Enter ID: 01
Book issued
```

5. **Return Book**

```
Enter choice: 5
Enter ID: 01
Book returned
```

6. **Delete Book**

```
Enter choice: 6
Enter ID: 01
Book deleted
```